

Configuring ABARS for Application Backup and Recovery

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Configuring ABARS

- Procedure
- Initialization Parameters
- High Level Qualifiers
- Management Class
- Aggregate Group
- Selection Data Sets
- RACF Considerations



Procedure

- Started task
- Initiated by DFSMSHsm primary address space
- Executed as a secondary address space
- Requires its own procedure
 - ✓ Often placed in SYS1.PROCLIB

```
//DFHSMABR PROC
//DFHSMABR EXEC PGM=ARCWCTL,
//          REGION=0M,TIME=1440
//MSYSOUT   DD   SYSOUT=*
//MSYSIN    DD   DUMMY
//SYSUDUMP  DD   SYSOUT=*
```

.....

.....

Procedure

- **Update RACF Started Class Table**
 - ✓ Define the ABARS procedure name
 - ✓ Associate it with the ABARS defined USERID
 - ✓ Performed by Security Administrator
- **ABARS commands are submitted to DFSMSHsm primary address space**
 - ✓ Batch TSO jobs
 - PGM=IKJEFT01
 - ✓ ISMF panel options
 - ✓ HSEND ABACKUP *agname* EXECUTE
 - ✓ HSEND ARECOVER *agname* EXECUTE

Initialization Parameters

- Reside in ARCCMDxx member of PARMLIB
- SETSYS commands are used to customize ABARS and set default values
 - ✓ ABARSPROCNAME
 - ✓ MAXABARSADDRESSSPACE
 - ✓ ABARSACTLOGMSGGLVL
 - ✓ ABARSACTLOGTYPE
 - ✓ ABARSDELETEACTIVITY
 - ✓ ABARSTAPES
 - ✓ ABARSUNITNAME

Initialization Parameters (continued)

- Define ABARS procedure name

SETSYS ABARSPROCNAME(DFHSMABR)

✓ Used by ABARS primary address space

- Limit number of concurrent ABARS address spaces

SETSYS MAXABARSADDRESSSPACE(nn)

✓ Often limited by number of available tape drives

✓ Controlled by ABARS primary address space

✓ Maximum number of concurrent aggregate backups and recovers

Initialization Parameters (continued)

- Activity log message level

SETSYS ABARSACTLOGMSGGLVL(FULL)

- ✓ ABARS activity logs only
- ✓ Full or reduced
- ✓ Recommend full

- Activity log data type

SETSYS ABARSACTLOGTYPE(DASD)

- ✓ DASD or SYSOUT
- ✓ Recommend DASD

Initialization Parameters (continued)

- **ABARS activity log disposition**

SETSYS ABARSDELETEACTIVITY(Y)

- ✓ Automatic deletion during ABARS roll-off processing

- **ABARS processing I/O buffer count**

SETSYS ABARSBUFFERS(5)

- ✓ Range of 1 (default) to 9
- ✓ Recommend 5

Initialization Parameters (continued)

- **ABACKUP output data set tape stacking option**
SETSYS ABARTAPES(STACK)
 - ✓ Write ABARS output files to minimum number of tapes
 - ✓ Default is one tape for each output file (2 to 4)
 - ✓ Recommend STACK
- **ABACKUP output data set tape unit name**
SETSYS ABARSUNITNAME(3490)
 - ✓ May also use esoteric names such as TAPEC
 - ✓ Esoteric names must also be specified in USERUNITTABLE
 - ✓ Can be overridden by ABACKUP command parameter

Initialization Parameters (continued)

- **ARECOVER ML2 output tape unit name**
SETSYS ARECOVERML2UNIT(3490)
 - ✓ Device type to be used for recovered ML2 data
 - ✓ Used at recovery site
- **Catalog status options for ABARS recovered GDSs**
SETSYS ARECOVERTGTGDS(SOURCE)
 - ✓ Options are:
 - **DEFERRED**
 - **ACTIVE**
 - **ROLLEDOFF**
 - **SOURCE (default)**

Initialization Parameters (continued)

■ Installation exits

SETSYS EXITON(BE)

- ✓ Aggregate backup error exit
 - ARCBEEEXT
- ✓ Aggregate backup ML2 bypass exit
 - ARCM2EXT
- ✓ Aggregate recovery bypass exit
 - ARCSKEXT
- ✓ Aggregate recovery conflict resolution exit
 - ARCCREXT

Initialization Parameters (continued)

- **ARECOVER DFDSS level 0 DASD target volume count**
SETSYS ABARSVOLCOUNT(ANY)
 - ✓ **ANY**
 - Use from 1 to 59 volumes for a recovered data set
 - ✓ **NONE**
 - Use same number of volumes used by source data set
- **Maximum ARECOVER utilization of non-SMS volumes**
SETSYS ARECOVERPERCENTUTILIZED(90)
 - ✓ **Default is 80%**

Initialization Parameters (continued)

- **DFSMSdss I/O processing option**
SETSYS ABARSOPTIMIZE(4)
 - ✓ Used when backing up level 0 DASD data sets
 - ✓ Recommend 4 => read 1 cylinder at a time

- **Display current ABARS parameters**
 - ✓ Issue HSEND QUERY ABARS command

High Level Qualifiers

■ Activity Logs

mcvtactn.Hmcvthost.ABACKUP.agname.Dyyddd.Thhmmss

mcvtactn.Hmcvthost.ARECOVER.agname.Dyyddd.Thhmmss

- ✓ Default for mcvtactn is HSMACT
- ✓ mcvthost supplied by DFSMSHsm procedure
- ✓ Used for both ABACKUP and ARECOVER activity logs
- ✓ Can be changed by using a patch
- ✓ Create a catalog ALIAS for the activity log HLQ
- ✓ Define as a functional RACF group
- ✓ Provide ALTER authority to the following:
 - ABARS procedure name
 - Storage Management staff
 - Technical Services staff (optional)
 - Others involved in data backup and recovery

High Level Qualifiers (continued)

■ ARECOVER processing

authid.agname.RESTART.Dyyddd.Thhmmss

authid.agname.CONFLICT.Dyyddd.Thhmmss

- ✓ authid defaults to HSM
- ✓ Create a catalog ALIAS for authid
- ✓ Define as a functional RACF group
- ✓ Provide ALTER authority to the following:
 - Storage Management staff
 - Technical Services staff (optional)
 - Others involved in data backup and recovery

Management Class

- Used to manage ABARS backup processing
 - ✓ Use ISMF to define or modify management classes
 - ✓ Specifications on ISMF panels 1-4 ignored (use defaults)
 - Panel 5 has all of the ABARS options
- Use a unique naming convention for ABARS
 - ✓ MCABARS
- Create as many as necessary, depending on options
 - ✓ Enqueue options (Continue or Fail)
 - ✓ Concurrent Copy/SnapShot requirements
 - ✓ Number of versions (cycles) to retain

MANAGEMENT CLASS DISPLAY

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Command ==>

CDS Name : MSC.SMS.SCDS
Management Class Name . . . : MCABARS

AGGREGATE Backup Attributes:

Versions : 1
Retain only Version . . :
Unit :
Retain extra Version . :
Unit :
Copy Serialization . . : CONTINUE
ABackup Copy Technique : STANDARD

Use UP Command to View previous Panel;
Use HELP Command for Help; Use END Command to Exit.

Management Class (continued)

- **Number of aggregate backup versions**
 - ✓ Maximum number of active backup 'generations' for each aggregate
- **Retention of last backup version**
 - ✓ Retention period for the last (only) backup version
 - ✓ Prevents deletion of last copy if backups for an aggregate are discontinued
 - ✓ Specify number of days, weeks, months, or years

MANAGEMENT CLASS DISPLAY

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Command ==>

CDS Name : MSC.SMS.SCDS
Management Class Name . . . : MCABARS

AGGREGATE Backup Attributes:

Versions : 5
Retain only Version . . : 1
Unit : YEAR
Retain extra Version . :
Unit :
Copy Serialization . . : CONTINUE
ABackup Copy Technique : STANDARD

Use UP Command to View previous Panel;
Use HELP Command for Help; Use END Command to Exit.



ISMF Management Class (continued)

- **Retain extra versions**
 - ✓ Retention period for non-current version
 - ✓ Prevents retention beyond practical usage period
- **Copy Serialization**
 - ✓ Action to be taken if a shared enqueue for a data set fails
 - **CONTINUE**
Fuzzy backup
 - **FAIL**
Abort

Management Class (continued)

- **ABACKUP Copy Technique**
 - ✓ Specify hardware assist backup option usage
 - **Concurrent Preferred**
Snapshot used if available
 - **Concurrent Required**
Backup fails if snapshot not available
 - **Standard**
Snapshot not used

MANAGEMENT CLASS DISPLAY

Page 5 of 5

Command ==>

CDS Name : MSC.SMS.SCDS
Management Class Name . . . : MCABARS

AGGREGATE Backup Attributes:

Versions : 5
Retain only Version . . : 1
Unit : YEAR
Retain extra Version . : 5
Unit : WEEKS
Copy Serialization . . : CONTINUE
ABackup Copy Technique : STANDARD

Use UP Command to View previous Panel;
Use HELP Command for Help; Use END Command to Exit.



Aggregate Group

- Defined using ISMF (option 9)
- Naming convention helps identify aggregate contents
 - ✓ Application name (acronym)
 - ✓ Backup cycle (D=daily, W=weekly, M=monthly, etc.)
 - ✓ Sequence number

For example: PAYD01, PAYD02, PAYW01, PAYM01, ...

- Number of concurrent backups to be created
 - ✓ Up to 15
- Management Class to be used

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Command ==>

CDS Name : MSC.SMS.SCDS

Aggregate Group Name : **PAYD01**

Description : PAYROLL APPLICATION BACKUP, DAILY, SET 1

Backup Attributes

Number of Copies : 2

Management Class Name . . : MCABARS

Output Data Set Name Prefix : ABARS.PAYD01

Account :

Use DOWN Command to View next Panel;

Use HELP Command for Help; Use END Command to Exit.

Aggregate Group (continued)

- Output data set name prefix
 - ✓ Use naming convention
 - Easily identify ABARS tapes
 - Used in vault pattern
 - Add aggregate name as part of the prefix

Example: ABARS.agname

Vault Pattern Description = ABARS.-.C01-

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Command ==>

CDS Name : MSC.SMS.SCDS

Aggregate Group Name : PAYD01

Description : PAYROLL APPLICATION BACKUP, DAILY, SET 1

Backup Attributes

Number of Copies : 2

Management Class Name . . : MCABARS

Output Data Set Name Prefix : ABARS.PAYD01

Account :

Use DOWN Command to View next Panel;

Use HELP Command for Help; Use END Command to Exit.

Aggregate Group (continued)

■ Selection Data Set(s)

- ✓ Sequential, PDS or PDSE data sets
 - Sequential recommended
- ✓ Up to 5 selection data sets
- ✓ Use a naming convention

ABARS.agname.SELECTn

ABARS.SELECT(agname)

■ Instruction data set

- ✓ Optional
- ✓ Seldom used
- ✓ Most installations use disaster recovery (DR) procedures written and maintained by all departments involved with recovery

AGGREGATE GROUP DISPLAY

Page 2 of 2
Command ===>

CDS Name : MSC.SMS.SCDS

Aggregate Group Name : PAYD01

To Browse a Data Set, Specify Number . . . (1, 2, 3, 4, 5, or 6)

Selection Data Sets

1 : **ABARS.PAYD01.SELECT1**

Member Name :

2 : **ABARS.PAYD01.SELECT2**

Member Name :

3 :

Member Name :

4 :

Member Name :

5 :

Member Name :

Instruction Data Set

6 : **ABARS.PAYD01.INSTRUCT**

Use UP Command to View previous Panel;

Use HELP Command for Help; Use END Command to Exit.

Selection Data Set

■ INCLUDE/EXCLUDE

- ✓ INCLUDE is a required parameter
- ✓ List of data sets to be backed up
- ✓ Cataloged DASD, migrated, and user tape data sets
- ✓ ACS type masking is supported

MY.DATA.SET.**

MY.*.SET.NAME

MY.**.NAME

MY.DA%%.SET.**

Selection Data Set (continued)

■ INCLUDE/EXCLUDE

✓ Generation Data Sets

MY.GDG.G0001V00	VALID
MY.GDG.G%%%%V%%	VALID
MY.GDG.*V01	VALID
MY.GDG(0)	VALID
MY.**	VALID
MY.GDG.G*	INVALID
MY.G*.*	INVALID

Selection Data Set (continued)

■ INCLUDE/EXCLUDE

```
INCLUDE(P216.P312522.TUNLDITE(0)
        P216.P312522.TUNLDITE(-1)
        P216.P312522.TUNLDITM(0)
        P216.P312522.TUNLDPID(0)
        P216.P316444.**
        P304.P2200008.DW220001(0)
        P305.P0050010.TMIFOUT(0))
EXCLUDE(P216.P316444.UNLOAD.*)
```

Selection Data Set (continued)

■ ALLOCATE/ALLOCATEEXCLUDE

- ✓ Optional parameter
- ✓ List of data sets to be allocated only (not copied)
- ✓ Cataloged DASD data sets only
- ✓ ACS type masking is supported

MY.DATA.SET.**

MY.*.SET.NAME

MY.**.NAME

MY.DA%%.SET.**

Selection Data Set (continued)

■ ACCOMPANY/ACCOMPANYEXCLUDE

- ✓ Optional parameter
- ✓ List of data sets to be cataloged only (not copied)
- ✓ Cataloged tape data sets only
- ✓ Typically used to catalog offsite tapes
- ✓ ACS type masking is supported

MY.DATA.SET.**

MY.*.SET.NAME

MY.**.NAME

MY.DA%%.SET.**

Selection Data Set (continued)

■ Miscellaneous

- ✓ Specific data set names can be in one list only
 - If in the INCLUDE list, cannot also be in the ALLOCATE or ACCOMPANY list
 - If INCLUDE list contains generic pattern names, must use EXCLUDE if they map to ALLOCATE or ACCOMPANY list names

INCLUDE(PAY.ME.FIRST.**)

EXCLUDE(PAY.ME.FIRST.ALWAYS.**)

ALLOCATE(PAY.ME.FIRST.ALWAYS.AND.OFTEN(0))

Selection Data Set (continued)

■ Miscellaneous

✓ Critical data set identification

- Rerun Recovery

Every input data set, in every batch job (JCL), for each critical application, is required to rerun the batch cycle at the DR site (*outputs are re-created during the rerun*)

- Forward Recovery

Every data set, both input and output, needed to process the next cycle at the DR site

- Some processing may also require outputs from other cycles

Previous day, week, month, quarter, or year

Selection Data Set (continued)

■ Maintenance

- ✓ Must be updated to reflect application changes
 - Add and delete data set names
 - Effort reduced if generic pattern names used
 - May back up unnecessary data

■ Typical reasons for application recovery failure

- ✓ #3 is lack of SDS maintenance
 - New data set names not added
- ✓ #2 is critical data sets not in the SDS
 - Missing data sets
- ✓ #1 is missing ABARS tapes
 - Not included in Tape Management vault list

RACF Considerations

- **STGADMIN Facility Class Profiles**
 - ✓ Protect against unauthorized command use
 - **ABACKUP and ARECOVER**
 - ✓ Can be generic or as granular as your site requires
- **Started Class Table**
 - ✓ Add the ABARS procedure name
 - ✓ Associate it with the ABARS defined USERID
 - ✓ Will likely require a Security Administrator unless you have RACF SPECIAL attribute

RACF Considerations (continued)

■ Generic example

- ✓ STGADMIN.ARC.ABACKUP.**
- ✓ STGADMIN.ARC.ARECOVER.**
 - Profile covers all aggregates defined
 - Permits authorized users to issue both commands
ABACKUP or ARECOVER
For all aggregates
 - Does not provide access to application data sets
Must have minimum of READ access to all
RACF protected data sets in aggregate group
 - No DFSMSHsm command authority required

RACF Considerations (continued)

■ Specific example

✓ STGADMIN.ARC.ABACKUP.PAYROLL

- Profile covers PAYROLL aggregate only
- Prevents use of ABACKUP command for unauthorized users
- Does not provide access to application data sets
 - Must have minimum of READ access to all RACF protected data sets in aggregate group
 - ALTER authority is required to use ABACKUP MOVE option
- No DFSMSHsm command authority required

RACF Considerations (continued)

- **Specific example**

- ✓ **STGADMIN.ARC.ABACKUP.PAYROLL**

- **Access recommendation**

- Job Scheduler**

- DR Staff**

- Technical Services**

- Storage Administration**

- Operations (optional)**

RACF Considerations (continued)

- **Miscellaneous**

- ✓ **Disaster Recovery Team**

- **Access recommendation**

- ABARS activity logs**

- ABARS RESTART and CONFLICT data sets**

- Application data sets (as required)**

**Thank you for attending SHARE
This is the end of**

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For

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